

WiDS Kalman Filtered Trend Trader

Assignment 1

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Question 1: Linear Regression

1. Model (scalar form). For $i = 1, \dots, n$,

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \varepsilon_i.$$

Model (Matrix form),

$$y = X\beta + \varepsilon.$$

Here,

y_i : observed response for observation i .

x_{ij} : observed value of predictor j (from 0 to p) for observation i (from 1 to n). By convention $x_{i0} = 1$ for the intercept(given).

$\beta_0, \beta_1, \dots, \beta_p$: unknown regression coefficients (parameters). β_0 is the intercept; β_j is the change in the mean response per unit change in x_j (holding other predictors fixed) under the linear model.

ε_i : error (noise, disturbance) for observation i . ε is the matrix form of the error.

2. Ordinary Least Squares (OLS) minimizes the SSE(Sum of squared errors / residuals), i.e., the squared difference between the observed responses y_i and the fitted values $\hat{y}_i$:

$$OLS \text{ minimizes } \sum (y_i - \hat{y}_i)^2$$

The MSE objective function for linear regression is:

$$MSE(\beta) = \frac{1}{n} \sum_{i=1}^n ((y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_p x_{ip})^2)$$

3. The MSE objective function in the matrix/vector form can be written as,

$$MSE(\beta) = \frac{1}{n} (\mathbf{y} - \mathbf{X}\beta)^T (\mathbf{y} - \mathbf{X}\beta)$$

$$MSE(\beta) = \frac{1}{n} (\mathbf{y}^T \mathbf{y} - 2\beta^T \mathbf{X}^T \mathbf{y} + \beta^T \mathbf{X}^T \mathbf{X} \beta)$$

$$\frac{\partial}{\partial \beta} MSE(\beta) = \frac{1}{n} (-2\mathbf{X}^T \mathbf{y} + 2\mathbf{X}^T \mathbf{X} \beta) = 0$$

$$-\mathbf{X}^T \mathbf{y} + \mathbf{X}^T \mathbf{X} \beta = 0$$

$$\mathbf{X}^T \mathbf{X} \beta = \mathbf{X}^T \mathbf{y}$$

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

4. $\mathbf{X}^T \mathbf{X}$ is invertible if and only if the columns of $\mathbf{X}$ are linearly independent (full column rank). Multicollinearity creates linear dependence among columns of $\mathbf{X}$. Linear dependence $\Rightarrow$ $\text{rank} < p+1 \Rightarrow \mathbf{X}^T \mathbf{X}$ becomes singular because $\text{rank}(\mathbf{X}^T \mathbf{X}) = \text{rank}(\mathbf{X})$. Hence, OLS cannot compute a unique $\hat{\beta}$.

5. Normal equations:

$\mathbf{X}^T \mathbf{X} \hat{\beta} = \mathbf{X}^T \mathbf{y}$, This is obtained on expansion of $(\mathbf{y} - \mathbf{X}\hat{\beta})^T (\mathbf{y} - \mathbf{X}\hat{\beta})$.

Residual orthogonality:

$$\mathbf{X}^T (\mathbf{y} - \hat{\mathbf{y}}) = 0$$

Hence proved.

- 6.

$$J(\beta) = \frac{1}{2n} |\mathbf{X}\beta - \mathbf{y}|^2$$

$$J(\beta) = \frac{1}{2n} (\mathbf{X}\beta - \mathbf{y})^T (\mathbf{X}\beta - \mathbf{y})$$

Taking gradient w.r.t. beta,

$$\nabla_{\beta} J(\beta) = \frac{1}{n} \mathbf{X}^T (\mathbf{X}\beta - \mathbf{y})$$

So, for batch gradient descent

$$\nabla_{\beta} J(\beta) = \frac{1}{n} \mathbf{X}^T (\mathbf{X}\beta - \mathbf{y})$$

$$\beta_{k+1} = \beta_k - \alpha \nabla_{\beta} J(\beta_k)$$

Where alpha is the learning rate,

$$\beta_{k+1} = \beta_k - \alpha \left(\frac{1}{n} \mathbf{X}^T (\mathbf{X}\beta_k - \mathbf{y}) \right)$$

7. In notebook

8. In notebook

9. In notebook

10. Violations in residual diagnostics indicate potential departures from the assumptions of linear regression. Heteroscedasticity suggests non-constant error variance, non-normal residuals affect inference validity, systematic patterns imply nonlinearity, and correlated residuals indicate dependence. These violations may result from model misspecification, omitted variables, or inappropriate functional form, and can often be addressed using transformations, robust methods, or alternative modeling approaches.

11. In notebook

12. Let the data be generated according to the model

$$y = f(x) + \varepsilon,$$

where:

$f(x)$ is the **true (unknown) regression function**.

ε is a random noise term satisfying:

$$\mathbb{E}[\varepsilon] = 0, \text{Var}(\varepsilon) = \sigma^2.$$

ε is **independent** of the training data and hence independent of the learned predictor $\hat{f}(x)$.

$\hat{f}(x)$ is the **estimator** learned from a random training sample; therefore $\hat{f}(x)$ is a random variable.

All expectations $\mathbb{E}[\cdot]$ are taken with respect to the joint randomness of the training data and the noise ε .

Objective:

We aim to derive the decomposition:

$$\mathbb{E}[(y - \hat{f}(x))^2] = \underbrace{(\mathbb{E}[\hat{f}(x)] - f(x))^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2]}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Irreducible noise}}$$

- **Step 1: Substitute the Data-Generating Model**

Using $y = f(x) + \varepsilon$, we write:

$$\mathbb{E}[(y - \hat{f}(x))^2] = \mathbb{E}[(f(x) + \varepsilon - \hat{f}(x))^2].$$

- **Step 2: Expand the Square**

$$(f(x) + \varepsilon - \hat{f}(x))^2 = (f(x) - \hat{f}(x))^2 + 2(f(x) - \hat{f}(x))\varepsilon + \varepsilon^2.$$

Taking expectation on both sides:

$$\mathbb{E}[(y - \hat{f}(x))^2] = \mathbb{E}[(f(x) - \hat{f}(x))^2] + 2\mathbb{E}[(f(x) - \hat{f}(x))\varepsilon] + \mathbb{E}[\varepsilon^2].$$

- **Step 3: Evaluate Each Term**

Noise term: By assumption

$$\mathbb{E}[\varepsilon^2] = \sigma^2$$

Cross Term: Since, ε is independent of $\hat{f}(x)$ and $\mathbb{E}[\varepsilon] = 0$,

$$\mathbb{E}[(f(x) - \hat{f}(x))\varepsilon] = \mathbb{E}[(f(x) - \hat{f}(x))]\mathbb{E}[\varepsilon] = 0$$

Thus, the cross term vanishes.

- **Step 4: Reduce the expression**

We now have:

$$\mathbb{E}[(y - \hat{f}(x))^2] = \mathbb{E}[(f(x) - \hat{f}(x))^2] + \sigma^2.$$

It remains to decompose $\mathbb{E}[(\hat{f}(x) - f(x))^2]$.

- **Step 5: Add and Subtract the Mean Predictor**

We write,

$$\hat{f}(x) - f(x) = (\hat{f}(x) - \mathbb{E}[\hat{f}(x)]) + (\mathbb{E}[\hat{f}(x)] - f(x))$$

- **Step 6: Square and Take Expectation**

Squaring:

$$\begin{aligned} (\hat{f}(x) - f(x))^2 &= (\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2 + (\mathbb{E}[\hat{f}(x)] - f(x))^2 \\ &\quad + 2(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])(\mathbb{E}[\hat{f}(x)] - f(x)) \end{aligned}$$

Now, taking expectation both sides,

$$\begin{aligned} \mathbb{E}[(\hat{f}(x) - f(x))^2] &= \mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2] + \mathbb{E}[(\mathbb{E}[\hat{f}(x)] - f(x))^2] \\ &\quad + 2\mathbb{E}[\mathbb{E}[\hat{f}(x)] - f(x)]\mathbb{E}[\hat{f}(x) - \mathbb{E}[\hat{f}(x)]] \end{aligned}$$

- **Step 7: Eliminate the Cross Term**

By the definition of expectation,

$$\mathbb{E}[\hat{f}(x) - \mathbb{E}[\hat{f}(x)]] = 0$$

Hence, the cross term vanishes.

- Step 8: Identify Bias and Variance

We obtain,

$$\mathbb{E} \left[\left(\hat{f}(x) - f(x) \right)^2 \right] = \underbrace{(\mathbb{E}[\hat{f}(x)] - f(x))^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2]}_{\text{Variance}}$$

- Step 9: Final Decomposition

Substituting back:

$$\boxed{\mathbb{E}[(y - \hat{f}(x))^2] = \text{Bias}^2 + \text{Variance} + \sigma^2.}$$

13. Given:

True Model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

Fitted model:

$$y = \alpha_0 + \alpha_1 x_1 + u$$

- a. From simple linear regression theory,

$$\alpha_1 = \text{Cov}(x_1, y) / \text{Var}(x_1)$$

Substitute the true model for y:

$$\text{Cov}(x_1, y) = \text{Cov}(x_1, \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon)$$

Using linearity of covariance and the assumption

$$\text{Cov}(x_1, \varepsilon) = 0:$$

$$\text{Cov}(x_1, y) = \beta_1 \text{Var}(x_1) + \beta_2 \text{Cov}(x_1, x_2).$$

Hence,

$$\boxed{\mathbb{E}[\alpha_1] = \beta_1 + \beta_2 \frac{\text{Cov}(x_1, x_2)}{\text{Var}(x_1)}}$$

(b) Show how omitting x_2 biases α_1

The bias is:

$$\text{Bias}(\alpha_1) = \mathbb{E}[\alpha_1] - \beta_1 = \beta_2 \frac{\text{Cov}(x_1, x_2)}{\text{Var}(x_1)}.$$

Interpretation

- If $\beta_2 \neq 0$ (i.e., x_2 affects y), and
- If $\text{Cov}(x_1, x_2) \neq 0$,

then the estimated coefficient α_1 is **biased**.

The sign of the bias depends on:

- the sign of β_2 ,
- the sign of $\text{Cov}(x_1, x_2)$.

b.

(c) When does the bias disappear?

The bias term is zero if either:

$$\beta_2 = 0 \quad \text{or} \quad \text{Cov}(x_1, x_2) = 0.$$

Meaning

- x_2 has no effect on y , or
- x_1 and x_2 are uncorrelated.

c.

14. So,

$$x_2 = x_1 + 0.9z, z \sim N(0,1)$$

This implies:

x_2 is almost the same as x_1

The predictors are highly correlated. This is multicollinearity.

a. The condition number of $X^T X$ is:

$$\kappa(X^T X) = \frac{\lambda_{\max}}{\lambda_{\min}}$$

- Large condition number $\rightarrow$ matrix is nearly singular
- Near singularity $\rightarrow$ numerical instability

Why does it become large here?

Because:

- Columns of X are almost linearly dependent
- One column adds almost no new information

(b) Why does variance of $\hat{\beta}$ increase?

OLS variance formula:

$$\text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$$

When predictors are highly correlated:

- $X^T X$ has very small eigenvalues
- Inverse has very large values

Result:

- Small noise $\rightarrow$ huge changes in coefficients
- Estimates become unstable

b.

(c) Why no bias, but high instability?

Key fact:

As long as:

$$\mathbb{E}[\varepsilon \mid X] = 0$$

OLS is unbiased:

$$\mathbb{E}[\hat{\beta}] = \beta$$

So:

- Multicollinearity **does NOT** shift the average estimate
- But it makes estimates very noisy

c.

(c) Why no bias, but high instability?

Key fact:

As long as:

$$\mathbb{E}[\varepsilon \mid X] = 0$$

OLS is unbiased:

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So:

- Multicollinearity **does NOT** shift the average estimate
- But it makes estimates very noisy

d.

